

MD. Tahidul Islam

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Profile Summary

Computer Science and Engineering Graduate from Pabna University of Science and Technology with a strong background in **Machine Learning**, **Deep Learning**, and Computer Vision. Passionate about research and development, with experience in Explainable AI, NLP, and AI-driven healthcare solutions. Proven track record in competitive programming, achieving **Pupil rank on Codeforces**, and active participation in AI and research conferences. **Skilled in Python, C++, and Java**, with expertise in web development and data analysis. Strong communication, leadership, and problem-solving skills demonstrated through research presentations and organizational involvement.

Education

B.Sc in Computer Science and Engineering <i>Pabna University of Science and Technology</i> ◦ CGPA: 3.66 ◦ Combined first at last semester.	<i>Pabna, Bangladesh</i> 2020 – 2025
Higher Secondary Certificate (H.S.C) <i>Tilna Degree College</i> ◦ Final Grade: 4.75	<i>Naogaon, Bangladesh</i> 2017 – 2019
Secondary School Certificate (S.S.C) <i>Tilna M.L. High School</i> ◦ Final Grade: 4.59	<i>Naogaon, Bangladesh</i> 2012 – 2017

Publications

- **Explainable Artificial Intelligence in Sensor-based Healthcare: Exploring Trends, Adoption Factors, Research Gap, and Future Research Directions**
AKM Bahalul Haque, Md. Safiul Islam, Md. Naimul Islam, **MD. Tahidul Islam**, Md. Kornel Ahmed Rahul
[Journal of Electronic Science and Technology-2025](#) 📄 Under Review
- **Undergraduate Thesis: The Impact of Large Language Models in Education: A Review of ChatGPT, DeepSeek, Gemini, and Qwen**
Supervisor:
Dr. Md. Abdur Rahim.

Conferences & Proceedings

- **A Novel Approach to ADHD Detection in Children with ASD via Handwriting-Based Machine Learning Models** 29 September 2025
MD. Tahidul Islam, Md Abu Johab, Md. Roton Ahmed, Nakib Aman
[2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking \(DOI: <https://doi.org/10.1109/QPAIN66474.2025.11171830>\)](#) 📄 Publisher: IEEE
- **Hybrid BiLSTM-BiGRU Model for Classification of Bangladeshi Dialects Using Textual Data** 18-19 July 2025

MD. Tahidul Islam, Md Abu Johab, Md. Roton Ahmed, Nakib Aman

[Springer International Conference on Data Science, AI and Applications - Gazipur, Bangladesh \(ICDSAIA-2025\)](#) [🔗 Published](#)(Not online yet)

- **The Key Determinants of Road Accidents: A Machine Learning Analysis** 18-19 July 2025

MD. Tahidul Islam, Md Abu Johab, Md. Roton Ahmed, Md. Nayem Hossain, Nakib Aman

[Springer International Conference on Data Science, AI and Applications - Gazipur, Bangladesh \(ICDSAIA-2025\)](#) [🔗 Published](#)(Not online yet)

- **EEG-Based Depression Detection Using CNNs and Heatmaps** 18-19 July 2025

Md Refat Sadiq, MD. Tahidul Islam, Most. Nur-A Marjan Esha, Md Mahmudul Hsan, Md. Abu Johab, Md. Roton Ahmed

[Springer International Conference on Data Science, AI and Applications - Gazipur, Bangladesh \(ICDSAIA-2025\)](#) [🔗 Published](#)(Not online yet)

- **CatBoost Classifier-Based Sadhu and Cholit Bangla Classification** August 28-30, 2025

MD. Tahidul Islam, Md Abu Johab, Md. Roton Ahmed, Nakib Aman

[3rd International Conference on Data Analytics and Insights – Kolkata, India \(ICDAI-2025\)](#) [🔗 Accepted](#)

- **Light Gradient Boosting Classifier for Real-Time Air Quality Monitoring** August 28-30, 2025

Md Abu Johab, MD. Tahidul Islam, Md. Roton Ahmed, Nakib Aman

[3rd International Conference on Data Analytics and Insights – Kolkata, India \(ICDAI-2025\)](#) [🔗 Accepted](#)

- **A Convolutional Neural Network-Based Model for Early Identification of Alzheimer Disease Using Structural MRI of the Brain** 27 September 2025

Sampad Kumar Singha, MD. Tahidul Islam, Saiful Islam, Nakib Aman, Md Anwarul Islam Himel

[3rd International Conference on Big Data, IoT and Machine Learning \(BIM 2025\)](#) [🔗 Accepted](#)

- **Breast Cancer Detection via Advanced Deep Learning Networks** 2025

MD. Tahidul Islam, Md Abdur Rahim, Abu Saleh Musa Miah, Md. Imran Hossain, Md. Shafiul Azam

[UCICS-2025](#) [🔗](#) (Published)

Projects

Intelligent Q&A Chatbot with Retrieval-Augmented Generation(RAG)

[GitHub](#) [🔗](#)

- Developed an intelligent Q&A chatbot using Retrieval-Augmented Generation(RAG) with LangChain, leveraging Ollama for embeddings and DeepSeekR1 via GroqAPI for response generation.
- Designed and implemented an interactive knowledge management system capable of ingesting PDFs and external sources, organizing information within a vector database, and providing accurate, context-aware responses through a Streamlit-powered interface.

Smart Diabetes Prediction using Machine Learning

[GitHub](#) [🔗](#)

- Developed a machine learning model to predict diabetes based on patient data. Utilized Python for implementation, leveraging Scikit-learn for model building and evaluation, and Pandas for efficient data prepro-

cessing and analysis. Focused on improving model accuracy through feature selection, data normalization, and cross-validation techniques.

- Tools Used: Python, Scikit-learn.

Artificial Neural Network - Digit Classification

[GitHub](#) 

- Designed a neural network in Python with TensorFlow to classify handwritten digits. Applied normalization and dropout to enhance performance and accuracy.
- Tools Used: Python, TensorFlow, Neural Networks

Nail Disease Detection Using LSTM/BiLSTM

[GitHub](#) 

- Developed a deep learning model for multi-class nail disease classification using a hybrid CNN and Bidirectional LSTM architecture. Extracted spatial features from images using convolutional layers and captured sequential dependencies through BiLSTM for improved classification accuracy across six categories. Applied data augmentation, class balancing, and evaluation techniques such as confusion matrix and F1-score.
- Tools Used: Python, Pytorch, Neural Networks.

Student Registration Form

[GitHub](#) 

- Developed a web-based student registration system using PHP and Laravel, integrated with MySQL for secure data storage. Designed responsive front-end interfaces with HTML and CSS to facilitate student enrollment, data validation, and record management.
- Tools Used: PHP, Laravel, MySQL, HTML, CSS

Technical Skills

Programming Languages: Python, C++, Java, C, MATLAB, Assembly Language (8085, 8086), Visual C++, Prolog

Machine Learning & AI: Supervised Learning, Unsupervised Learning, Reinforcement Learning, CNNs (Expert), RNNs, Computer Vision, NLP

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, NLTK, SpaCy, PyCaret

Web Technologies: HTML, CSS, Bootstrap, PHP, Laravel, .NET Core

Database & Tools: SQL, MySQL, Git, LaTeX, Linux, Windows

Achievements & Competitions

ICPC Asia Dhaka Regional Contest Participant 2024

- Onsite participant representing Pabna University of Science and Technology
- Hosted by Daffodil International University

2nd Place - Inter Department Individual Programming Contest 2024

- Pabna University of Science and Technology

Runner-up & 2nd Runner-up - Intra University Programming Contest 2023

- Multiple achievements in university-level programming competitions
- Pabna University of Science and Technology

17th Place - Inter University Programming Contest 2023

- North Western University

Problem Solving Profile

Codeforces: Solved 900+ problems, Max rank: Pupil (1243)

LeetCode: Solved 180+ problems

CodeChef: Solved 100+ problems

Other Platforms: AtCoder (20+), CSES (30+), LightOJ (30+), Others (20+)

Leadership & Volunteering

Founder

2024 – Present

Kishore Sangha

- Founded and leading youth organization
- [Facebook Page](#) 

Executive Member

2022 – 2024

Solver Green, Pabna University of Science and Technology

- Active member of environmental and sustainability club
- [LinkedIn Company Page](#) 

Member

2019 – Present

Sapahar Maun Club

- Long-term member of community service organization
- [Facebook Page](#) 

Languages

Bangla: Native

English: B2 (Listening), B1 (Reading), C1 (Writing), A2 (Speaking)

Interests

Football, Badminton, Chess, Cricket, Carrom, Gardening

References

Dr. Md. Abdur Rahim

Associate Professor & Chairman

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Nakib Aman

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